



LINEAR ALGEBRA

UE19MA251

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Agenda – Problems on SVD



- Example 1

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Example 1. Find the singular value decomposition for the matrix:

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 3 & 2 & -2 \end{bmatrix}$$

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Agenda – Problems on SVD



Solution: The singular value decomposition

of A is given by $A = U \Sigma U^T$

2×3 2×3 1×3 2×3

where $U = A A^T = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$, a square symmetric matrix.

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Let $B = AA^T$. Eigenvalues of B
can be found from: $|B - \lambda I| = 0$

$$\therefore \lambda_1 = 25, \lambda_2 = 9$$

Corresponding Eigenvectors are:

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

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Agenda – Problems on SVD



Note that x_1 and x_2 are orthogonal.

The orthonormal vectors are

$$u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \text{and} \quad u_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\therefore U = [u_1 \ u_2] = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

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Agenda – Problems on SVD



Singular values are the square roots
of the eigenvalues.

$$\therefore \sigma_1 = \sqrt{\lambda_1} = \sqrt{25} = 5$$

and $\sigma_2 = \sqrt{\lambda_2} = \sqrt{9} = 3$

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Agenda – Problems on SVD

Define $v_i = \frac{u_i^T A}{\sigma_i} ; i = 1, 2$

$$\therefore v_1 = \frac{u_1^T A}{\sigma_1} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}^T$$

$$v_2 = \frac{u_2^T A}{\sigma_2} = \begin{bmatrix} -1/\sqrt{32} & 1/\sqrt{32} & -4/\sqrt{32} \end{bmatrix}^T$$

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Agenda – Problems on SVD

$$\tilde{V} = [v_1 \ v_2 \ v_3] = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{32} & * \\ 1/\sqrt{2} & 1/\sqrt{32} & * \\ 0 & -4/\sqrt{32} & * \end{bmatrix} \quad \begin{matrix} \\ \\ v_3 \end{matrix}$$

The 3rd vector v_3 must be orthogonal
to both v_1 and v_2 .

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Agenda – Problems on SVD

$$\therefore \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{32} & 1/\sqrt{32} & -4/\sqrt{32} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving, we get $\frac{v_3}{\|v_3\|} = \frac{1}{\|v_3\|} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$

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Agenda – Problems on SVD

Thus $v^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{32} & 1/\sqrt{32} & -4/\sqrt{32} \\ -2/3 & 2/3 & 1/3 \end{bmatrix}$

The SVD is

$$A = U \Sigma v^T$$

where $\Sigma = \begin{bmatrix} \sqrt{25} & 0 & 0 \\ 0 & \sqrt{9} & 0 \end{bmatrix}_{2 \times 3}$



THANK YOU

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